

Article

From Point Clouds to FEM: A Practical Framework for the Automated Modeling of Transmission Towers

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Abstract

Finite element modeling is central to the safety assessment and structural health monitoring of transmission towers, yet conventional workflows remain labor-intensive and depend on complete engineering drawings, which are often unavailable for aging towers. This study proposes an automated finite element modeling framework based on unmanned aerial vehicle (UAV) point clouds. Multi-view stereo reconstruction with geometric-photometric depth refinement is used to generate dense three-dimensional point clouds, from which key nodal coordinates are extracted and converted into executable ANSYS APDL command streams. Material properties, boundary conditions, and loads are then assigned automatically to construct computable finite element models. This method was first validated using concrete cubes and Q235B steel members and was further evaluated on a scaled transmission tower through comparison with the commercial SmartTower workflow. The results show that the proposed approach reduces modeling time from 1.5 days to 4.3 h, corresponding to an 88% efficiency improvement, while keeping nodal coordinate errors within 10% and modal frequency errors within 3.7%. The framework satisfies the basic requirements of transmission tower numerical analysis and provides a practical non-contact solution for the rapid assessment of aging towers lacking design documentation.

Keywords: transmission tower; structural health monitoring; computer vision; point cloud model reconstruction; finite element modeling

1. Introduction

As critical infrastructure components of modern power grids, transmission towers play a vital role in ensuring reliable electricity transmission across long distances while withstanding extreme environmental loads [1]. Statistical reports indicate the length of 220 kV+ voltage level transmission lines in China rising from 5.439 billion km to 9.197 billion km between 2013 and 2023 (69% growth) [2]. The magnitude of this grid expansion proportionally increases the operational and maintenance burdens on infrastructure engineers, particularly for aging tower networks. However, these towers are often located in remote and harsh environments, where they are susceptible to multiple environmental stressors, such as high winds, lightning strikes, ice loads, corrosion, and material degradation caused by mechanical fatigue [3,4]. These factors can lead to the gradual accumu-

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lation of structural damage, and if these damages are not detected and addressed promptly, they may reduce the load-bearing capacity of the tower and even lead to structural failure. Thus, structural health monitoring (SHM) and timely maintenance of transmission towers are especially crucial [5–8].

Traditional SHM for transmission towers predominantly employs a dual approach combining physical sensor networks and finite element analysis (FEA) [9]. Sensor deployment techniques involve placing sensors at critical points of the tower to collect real-time deformation and displacement responses, thereby reflecting the actual state of the structure and identifying localized issues. Meanwhile, the FEA of transmission towers can assess the overall structural behavior. Ultimately, by integrating the strengths of both methods, the monitoring system is capable of performing comprehensive modal identification and reliability assessments of transmission towers. Nonetheless, for large structures such as transmission towers that frequently carry high-voltage electricity during operation, physical sensor networks face significant challenges including difficult installation, high costs, limited measurement points, and restricted monitoring durations [10]. More critically, FEA implementation encounters two fundamental limitations: (i) labor-intensive model construction requiring 5–7 days per tower—increasingly inadequate for growing inspection demands—and (ii) mandatory reliance on as-built drawings. Industry reports and our fieldwork confirm that a large number of the aging towers in the areas studied lack complete drawings.

In recent years, with the rapid development of image processing technology, computer vision-based monitoring methods have emerged as a promising alternative [11,12]. With advantages such as long-range capability, non-contact measurement, high precision, and time efficiency, these methods address many limitations of traditional approaches and have increasingly gained favor among engineers and researchers [13–17], gradually finding application in practical engineering projects. Computer vision-based monitoring systems typically incorporate techniques such as feature detection, feature extraction, feature matching, and feature tracking, analyzing digital image and video data to assess structural conditions [18]. And they have shown promising capabilities in areas such as displacement time-history signal acquisition [19–21], structural crack detection [22], and deformation damage assessment [23]. Tang et al. [24] proposed a method for the real-time, continuous monitoring of the structural posture of long-span bridge cables. The feasibility of this approach was verified through theoretical simulations, laboratory tests, and practical applications. Zhu et al. [25] proposed a dual-camera-based visual monitoring system to address displacement monitoring during the construction and post-construction phases of prefabricated subway stations, and this system has been implemented in real engineering projects. Zhu et al. [26] developed a computer vision-based structural vibration identification system using techniques such as image preprocessing, phase optical flow estimation, mode decomposition, and phase motion amplification, successfully detecting structural vibrations in complex environments. Experimental and field test results indicate that this system can accurately detect structural vibrations in complex environments.

However, the overall technical approach of the visual monitoring systems developed by researchers has primarily focused on the collection of real-time monitoring data, with limited contribution to the establishment of structural mechanical models. For engineering structures such as transmission towers, comprehensive and real-time health monitoring is essential, and FEM remains the standard method for addressing global analysis challenges. Therefore, exploring the integration of traditional FEM with computer vision-based monitoring systems is of significant importance.

Since the beginning of the 21st century, with the rapid advancement of computational capabilities, computer vision has achieved revolutionary progress, which also serves as an inspiration for the methods proposed in this paper. Researchers have started

integrating computer vision with machine learning, deep learning [27–29], UAVs [30], and other advanced algorithms to explore increasingly complex visual monitoring tasks, including 3D reconstruction [31–35] and motion analysis. Liu et al. [36] developed a computer vision-based framework that extracts 1D vibration signals from edge pixels in video footage and, using a 1D convolutional neural network (CNN) and long short-term memory (LSTM) network, derives structural modal parameters from these signals. Khuc et al. [37] proposed an enhanced non-contact displacement measurement method that combines UAVs and computer vision algorithms to address challenges in traditional visual measurement methods when dealing with high or inaccessible areas. As for studies on 3D reconstruction, Qu et al. [38] proposed a method for obtaining the 3D coordinates of the surface feature points of architectural structures based on computer vision. This method reconstructs the target structure by utilizing camera motion and tests its feasibility on seven different buildings. Lin et al. [39] proposed an automated high-precision geometric reconstruction method for cable net structures, which includes extracting cable shapes and forces from laser-scanned point clouds. This method determines the shape and force of cables by extracting the coordinates of key cable regions, applying segmented chain theory and nonlinear optimization, and was validated on single-curvature cable net structures. Angelosanti et al. [40] introduced a new workflow that directly converts magnified digital image correlation (M-DIC) point cloud data into surfaces within building information modeling (BIM) systems. This approach addresses the challenge of efficiently managing and visualizing large-scale dynamic SHM data.

To the best of the authors' knowledge, existing visual monitoring systems primarily focus on improving the accuracy of real-time measurement data and the creation of virtual 3D models, while relatively little research has been conducted on the integration of visual technologies with FEM. To overcome the limitations of traditional FEM methods, which are heavily reliant on engineering drawings, time-consuming, labor-intensive, and complex to operate, this study proposes an automated FEM approach for transmission towers via UAV-Based Point Clouds. The proposed approach utilizes advanced multi-view vision principles to reconstruct the target structure's point cloud model through UAV-captured multi-angle images, achieving over 95% feature coverage. Subsequently, depth-aware densification through geometric photometric optimization yields a dense point cloud with substantially higher number of points and accuracy. Finally, based on the point cloud model, the automated APDL (ANSYS Parametric Design Language) command stream generator developed by the research team is used to systematically connect the key nodes and assign material properties, loads, and boundary conditions, which are then integrated into the ANSYS APDL command stream to achieve the automated finite element modeling of transmission towers. This approach reduces labor and resource consumption, significantly enhances modeling efficiency, and aims to establish an FEM using computer vision technology. Section 2 of this paper provides a detailed explanation of the principles of the proposed method, while Sections 3 and 4 experimentally verify the feasibility and applicability of the method. Section 5 provides a summary of the entire paper and offers an outlook on the future development of this field.

2. Materials and Methods

2.1. Basic Steps

The FEM approach to the automation of transmission towers proposed in this study is centered on the point cloud model reconstruction technology. As shown in Figure 1, this method comprises three main components: the collection and processing of image datasets, sparse and dense point cloud reconstruction, and command stream generation for FEM.

Firstly, a UAV is used to capture images of the target structure from various angles, followed by image processing to obtain an unordered image dataset of the structure. Subsequently, feature detection and matching are performed on the image dataset. Based on the registration results, triangulation, pose estimation, and model optimization are conducted to generate a sparse point cloud model of the structure. Following this, based on multi-view stereo (MVS) principles, the depth information from the image dataset is utilized to fill and fit the sparse model using optical and geometric consistency, thereby completing the dense point cloud reconstruction. Finally, the coordinates of the dense point cloud model are transformed into actual coordinates using the rigid body transformation law, which are then integrated with FEM software (ANSYS 18.0) for the FEM of the structure. This novel approach successfully converts virtual three-dimensional models established through visual technology into mechanical models of engineering structures.

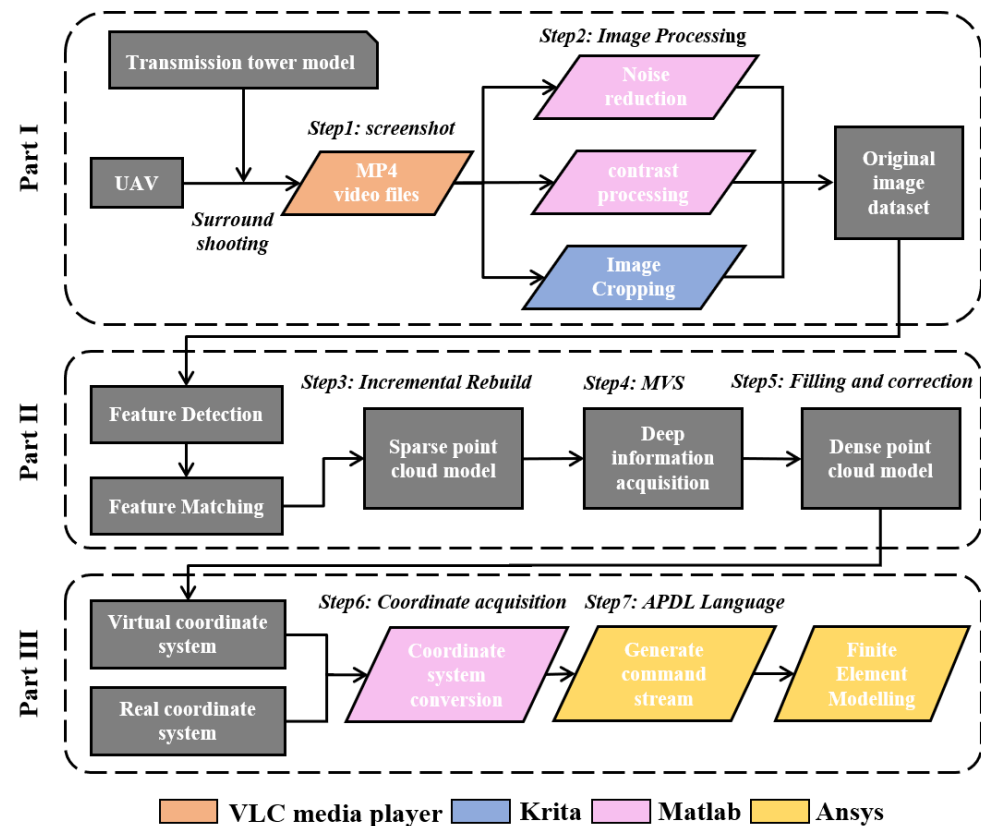


Figure 1. Flowchart of FEM method for transmission towers based on point cloud model reconstruction.

2.2. Image Acquisition and Processing

The collection of image datasets is a critical stage in the research process, directly influencing the accuracy and quality of the final point cloud model. In this study, the original image set consists of video frames captured by a camera mounted on a UAV. This method ensures that all details of the structure's surface are captured, thereby enhancing the completeness and reliability of the point cloud model and significantly saving labor and resources. Additionally, to ensure the accuracy of the FEA of the final structure, it is essential to capture images with consistent camera parameters. Therefore, disabling features such as autofocus, automatic FPS, and auto white balance during shooting is also an important step.

Considering that reconstructing a point cloud model is often computationally intensive, it is critical to maintain the coordinate accuracy of numerous point sets. This study

proposes that the careful processing of the original image set can potentially address this challenge. Cropping images along the structure's contours minimizes unnecessary feature point detection and matching, significantly improving computational efficiency. Additionally, in computer vision, lighting and noise often affect the accuracy of feature detection; thus, preprocessing operations such as noise reduction and contrast enhancement are necessary to optimize the image set.

2.3. Point Cloud Model Reconstruction

Point cloud model reconstruction is the most critical stage in performing the FEM of civil engineering structures using computer vision systems. This process utilizes the principles of “pinhole imaging” and multi-view vision to convert real 3D coordinates to planar 2D coordinates and recover the target structure in a computer-constructed virtual coordinate system. After obtaining the image dataset, this stage primarily comprises sparse point cloud reconstruction, depth information acquisition, and dense point cloud reconstruction. A schematic of the point cloud model reconstruction process is shown in Figure 2.

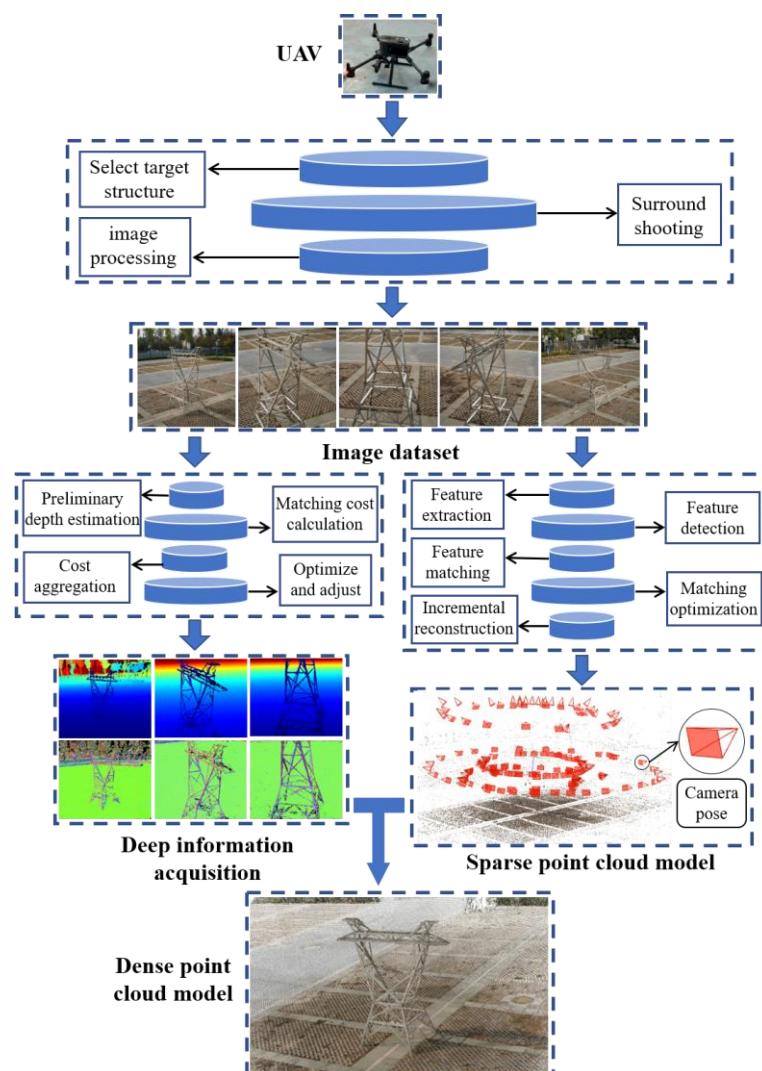


Figure 2. Schematic diagram of point cloud reconstruction process for transmission towers.

2.3.1. Constructing Sparse Point Cloud Model of Transmission Tower

In the sparse point cloud reconstruction module, the main approach adopted in this study is to incrementally add images for feature detection and matching, iteratively performing the 3D reconstruction of the target structure. Simultaneously, the camera poses are estimated based on the reconstructed model, and a global optimization method is applied to continuously improve model accuracy until all usable images in the dataset are fitted. The final output includes all camera parameters and a sparse point cloud model that provides the basis for subsequent dense point cloud reconstruction.

Given the “multi-corner, multi-edge” characteristics of transmission tower structures, this study selects the Scale-Invariant Feature Transform (SIFT) algorithm for feature detection. This algorithm, with its robust scale and rotation invariance, is particularly suited for image matching and 3D reconstruction in computer vision applications. During feature detection and extraction, the computer identifies feature points of the transmission tower structure based on specified recognition parameters. To enhance the accuracy of point cloud reconstruction, appropriate peak and edge response thresholds must be adjusted, although this may impact computational speed. Additionally, the detected feature points include RGB values, which improve the realism of the point cloud model and facilitate the selection of key node locations for constructing the FEM. Moreover, EXIF files within the image set, containing the camera’s internal parameters such as focal length, aperture, and shutter speed, are also extracted to prepare for subsequent steps.

Next, the detected and extracted feature points are matched. Given the large number of feature points, this study uses the “nearest neighbor search algorithm” to accelerate matching between image datasets. Symmetric matching and the RANSAC algorithm are applied to filter and optimize the matched pairs, effectively improving the accuracy and reliability of the matching process. Symmetric matching refers to bidirectional matching between two images: first matching from Image A to Image B and then from Image B to Image A. Only feature pairs that match successfully in both directions are retained. The retained matching pairs are then subjected to geometric verification to ensure they meet geometric constraints. After matching is completed, a feature point matching matrix is generated. The displayed image is a heatmap of the matching results, where colors range from dark (red) to light (blue), representing confidence levels from high to low. This heatmap reflects, to some extent, the density of feature matching between image sets as well as the accuracy of the point cloud model.

In fact, the previous steps only completed the initial matching by selecting the highest-quality image pair from the dataset as a foundation, without utilizing the full information from the entire dataset. To complete the initial sparse point cloud model, incremental reconstruction is required. As shown in Figure 3, using the initial matched pair as the base, additional images are gradually added to the reconstruction process. For each new image added, feature matching and 3D point cloud updates are performed. Meanwhile, using the previously obtained internal camera parameters and the initial camera pose estimation, the Perspective-n-Point (PnP) algorithm is applied to estimate the camera pose of the new image. This is an iterative updating process, where the continuous addition of images and viewpoints iteratively improves and refines the sparse point cloud model. Unlike the previous feature matching validation process between two images, incremental reconstruction requires multi-image matching validation with loop closure verification. This ensures that feature point matches in the image sequence form a consistent loop, thereby enhancing the reliability of the matches.

Finally, this study further optimizes and refines the generated point cloud model. Through statistical and geometric validation, incorrectly matched feature points and misestimated 3D points were removed, and the accuracy of the camera pose and 3D point

coordinates in the model was improved to obtain a highly accurate sparse point cloud model.

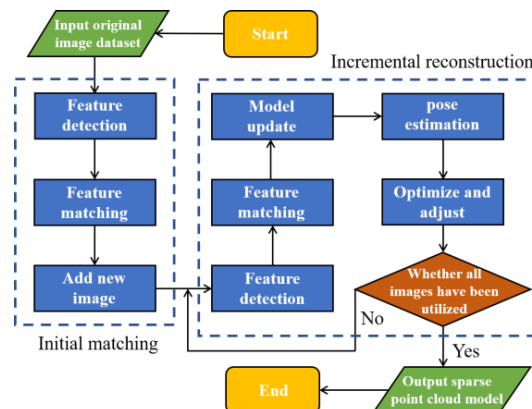


Figure 3. General procedure for incremental reconstruction of point cloud models.

At this stage, the sparse point cloud reconstruction of the transmission tower structure is complete, providing numerous feature points on the tower body and the camera pose for each measurement point. However, to apply this to practical engineering and bridge the gap between computer vision and structural analysis, it is necessary to further construct a dense point cloud model to enhance model accuracy.

2.3.2. Getting Depth Information from Image Dataset

The sparse point cloud model contains only a limited number of 3D points and may have considerable gaps and incomplete regions. To accurately describe the surface details of the structure and transition to a dense point cloud, depth information serves as a crucial bridge. Depth information here refers to the distance between each pixel and the camera. This study employs an MVS reconstruction approach, primarily involving disparity calculation, matching cost computation for corresponding features on reference images, cost aggregation, and optimization to ultimately obtain depth information. However, before this process, the original image set must be undistorted, as skipping this step would introduce significant errors.

Initial depth estimation is first conducted using feature point matching results, applying geometric constraints and photometric consistency to obtain preliminary depth information. Geometric constraints ensure the spatial relationships of feature points across different viewpoints, while photometric consistency ensures that corresponding points have similar color or brightness. The basic principle of disparity calculation is used here to convert disparity into depth information. Disparity represents the positional difference in the same object across images from different viewpoints and is inversely related to depth information. The calculation process is shown in Equations (1) and (2):

$$Disparity = x_1 - x_2 \quad (1)$$

In Equation (1), x_1 and x_2 represent the pixel coordinates of the same point in different images. According to the disparity formula, the depth is calculated as follows:

$$Depth = \frac{f \cdot B}{Disparity} \quad (2)$$

In Equation (2), f represents the focal length of the camera, and B is the baseline distance between the two cameras.

Secondly, since the initial depth estimates are often rough and contain significant errors, further optimization and refinement of the depth information is necessary. This requires calculating the matching cost for each pixel to assess the similarity and matching

degree of corresponding pixels between images. For each pixel (u, v) , the cost function can be expressed as follows:

$$C(u, v, d) = \sum_{(u', v') \in N(u, v)} |I_1(u', v') - I_2(u' - d, v')| \quad (3)$$

In Equation (3), d represents the disparity, and $N(u, v)$ denotes the neighborhood of pixel (u, v) . The cost function is then aggregated within a certain range to reduce noise and incorrect matches:

$$C'(u, v, d) = \sum_{(u', v') \in W(u, v)} C(u', v', d) \quad (4)$$

In Equation (4), $W(u, v)$ represents the window around pixel (u, v) . Based on the aggregated cost function, the position with the minimum cost is selected as the depth value of the pixel:

$$d(u, v) = \arg \min_d C'(u, v, d) \quad (5)$$

Finally, the calculated depth information is optimized and refined to eliminate the effects of noise and discontinuities, resulting in the final depth information. The computer also integrates “photometric consistency” and “geometric consistency” to generate high-precision depth maps and normal vector maps. For photometric consistency, the computer uses lighting information from the image set (such as pixel color and brightness) to determine the correspondences of the same physical point from different perspectives. However, relying solely on photometric consistency is prone to errors due to variations in lighting, shadows, and reflections and a lack of texture. In contrast, geometric consistency depends on the position and structure of points in 3D space and is unaffected by lighting, reflections, or texture, providing more robust matching. The complementary nature of these two methods enables the computer to achieve optimal depth information.

At this stage, by combining the detection of surface feature points from the sparse point cloud model with the acquisition of depth information for all pixel points in the image set, the conditions are met to construct a dense point cloud model that more closely represents the actual structure.

2.3.3. Constructing Dense Point Cloud Model of Transmission Tower

In order to construct a high-quality dense point cloud model, after the first few steps of obtaining depth information from the image dataset, the steps of depth map fusion, filling, and optimization are required to obtain the optimal solution of the fitted surface of the densely filled sparse point cloud. Depth map fusion involves aggregating depth maps from multiple viewpoints to generate a unified point set that represents the 3D structure of the entire scene. During the aggregation process, overlapping regions from multiple viewpoints are depth-fused, selecting the most reliable depth values to form an optimal point set that includes as much scene detail as possible. Next, based on the original sparse point cloud spatial coordinate system, the aggregated optimal point set is incorporated into the previously generated sparse point cloud model and corrected using an error correction algorithm.

At this point, a dense point cloud model of the transmission tower structure is successfully constructed. This virtual model will be used for the FEM of the engineering structure.

2.4. Automated Finite Element Modeling Based on Point Cloud Models

In civil engineering, the construction and analysis of FEM structures involve establishing a mechanical model and computing and integrating discrete elements. Taking a transmission tower as an example, engineers typically establish a mechanical model based on the actual structure of the tower and use modeling tools to create the corresponding

structural model. They then use software like Abaqus 2023, ANSYS 18.0, or HyperMesh 2022 to divide the computational object into multiple numbered elements for FEA. The novel FEM method proposed in this study significantly accelerates the process of constructing the mechanical model. The help of a dense point cloud model provides precise coordinate information on key transmission tower nodes. This study uses ANSYS FEA software (ANSYS 18.0) as an example and develops an automated APDL command stream generator. The developed algorithm takes nodal data extracted from the point cloud model as input and systematically connects these nodes to construct the structural framework. Crucially, the generator integrates a comprehensive material property database—a 15-year curated repository by our research team—containing standardized parameters for transmission tower components, boundary conditions, and load configurations [41–50]. This database enables rapid parameter assignment to the generated framework, with the system ultimately exporting executable APDL commands in TXT format.

It should be emphasized that material properties are not derived from the point cloud itself but are assigned from a predefined database of standard transmission tower materials and sections. The vision-based inference of mechanical properties remains beyond the scope of this study.

To demonstrate the contribution of the proposed method to the mechanical modeling of transmission towers, this study adopted a two-phase validation approach. Initially, feasibility assessment tests were conducted on fundamental concrete blocks and steel structural specimens to verify methodological validity. Subsequently, the developed FEM scripting system was applied to establish a scaled transmission tower model (Figure 4), where the automatically generated APDL commands were executed in ANSYS for numerical simulation. Modal analysis served as the representative case study to evaluate modeling performance, with comprehensive experimental procedures and the analytical results detailed in Sections 3 and 4.



Figure 4. Transmission tower scale-down model.

The finite element modeling process of the transmission tower is illustrated in Figure 5. The target structure of Figure 4 contains many of the desired critical nodes, though only a subset of these points are identifiable in the sparse point cloud model (Figure 5a). By contrast, the dense point cloud model (Figure 5b) accurately captures the positions of these nodes and extracts their coordinates. Using the developed script, these nodes are automatically and systematically connected to form the structural framework (Figure 5c). Finally, material properties, boundary conditions, and loads are assigned to generate the complete finite element model (Figure 5d).

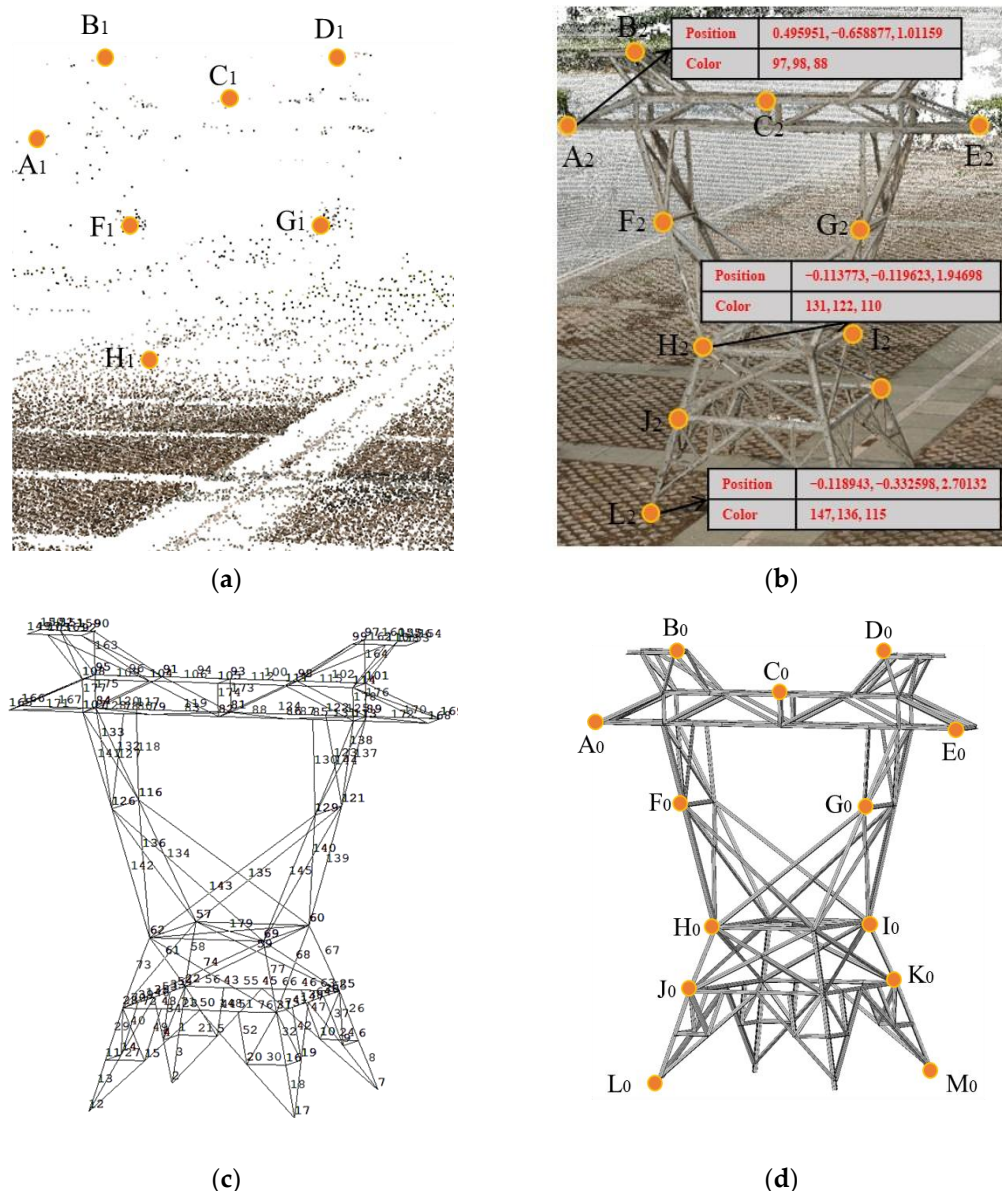


Figure 5. Stages of constructing transmission tower FEM: (a) sparse point cloud model and some nodes; (b) dense point cloud model and some nodes; (c) generated structural framework; (d) FEM of transmission tower structure.

3. Benchmark Testing of Automated FEM on Basic Structural Specimens

In this study, benchmark tests on simple concrete and steel specimens are first carried out to systematically verify the applicability of the proposed modeling method in foundation components. Concrete specimen blocks (C30, 200 mm cube) and Q235B steel I-sections (I-beam, 350 × 150 × 30 × 18 mm) are selected as the test objects.

The test process mainly consists of the following (Figures 6 and 7 illustrate the test flow and details): (1) A smartphone (iPhone 15 (Apple Inc., Cupertino, CA, USA)), frame rate: 60FPS, resolution: 4K) is used to capture the image data of the target structure at all angles. (2) The captured image dataset is preprocessed in MATLAB 2023a to mitigate the effects of noise and illumination on the test. (3) Through the above process in this paper, the sparse point cloud model and dense point cloud model of the two structures are constructed. (4) Based on the point cloud model, a finite element model is generated by the developed automated APDL command stream generator.

It should be noted that no auxiliary equipment (e.g., tripod, rail, calibration target) was used during smartphone image acquisition for the benchmark tests. To mitigate human-induced geometric deviations, the capture process followed a structured multi-angle trajectory with >50% image overlap, and all frames were subsequently cropped and de-noised prior to reconstruction.

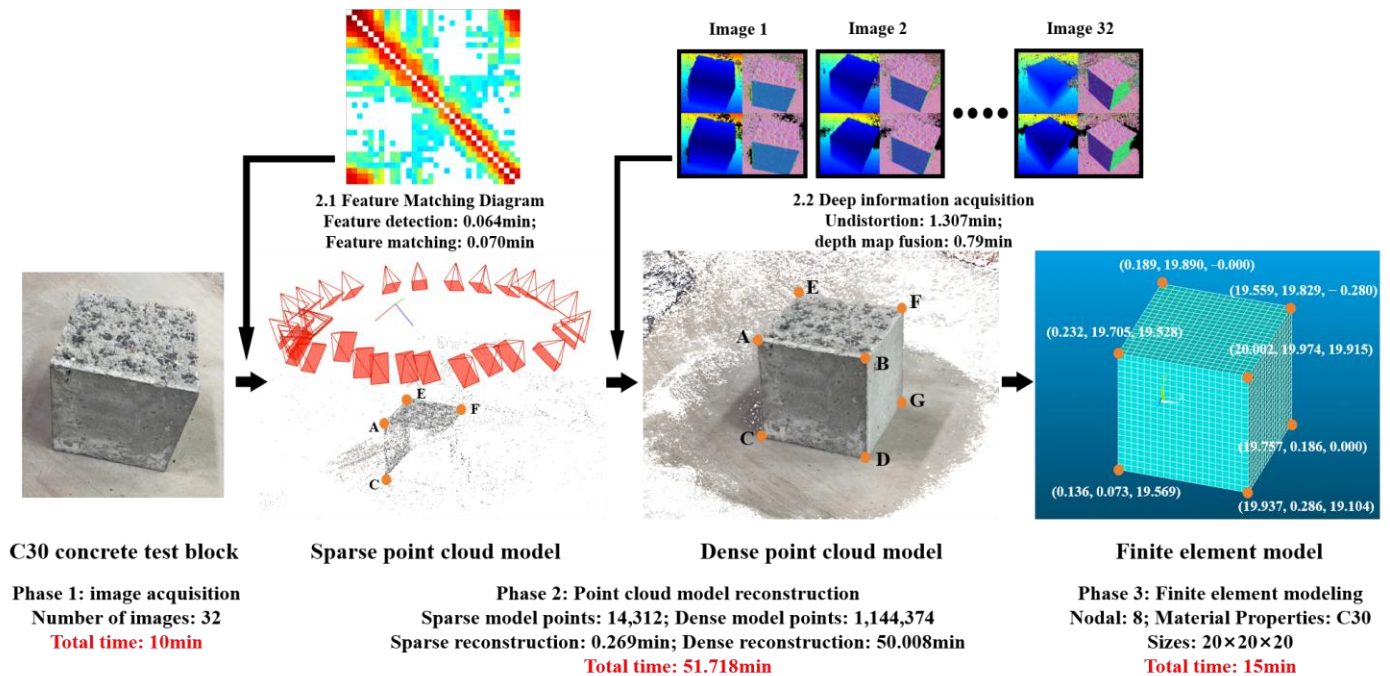


Figure 6. Workflow of concrete cube testing via point cloud FEM.

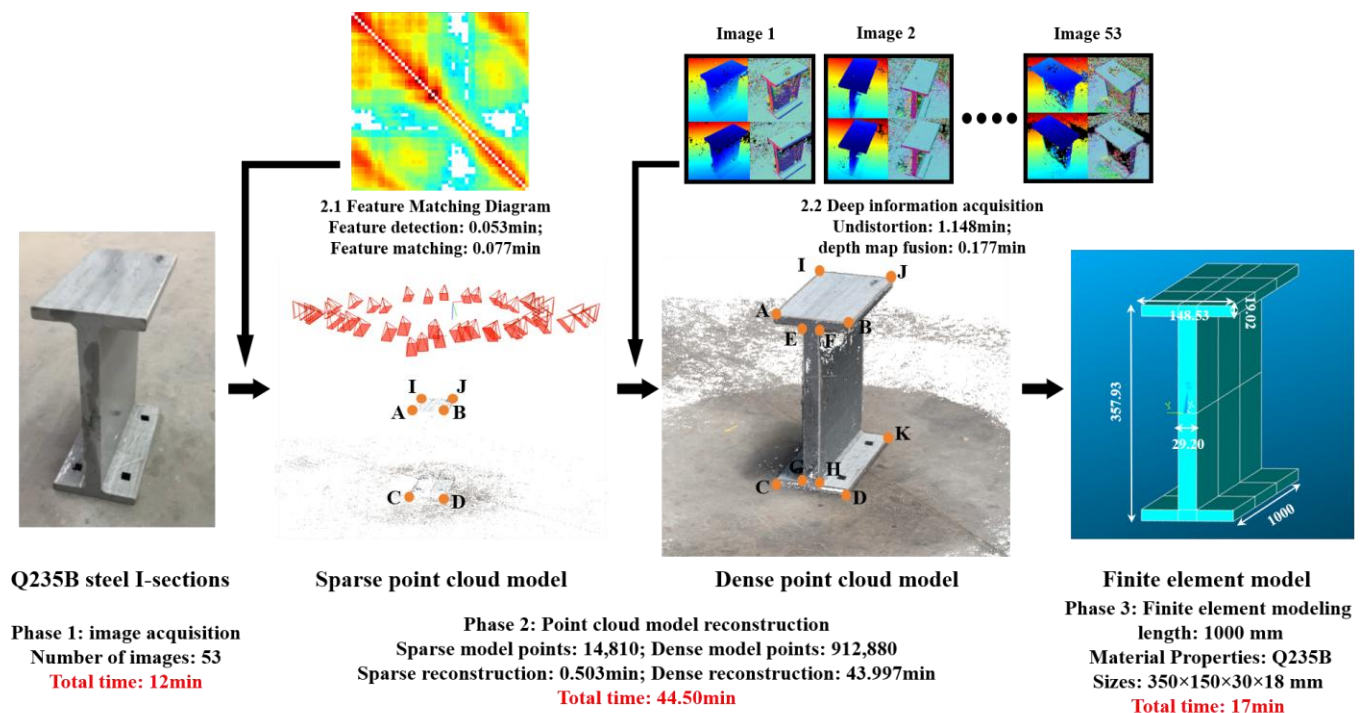


Figure 7. Workflow of Q235B steel I-section testing via point cloud FEM.

The test results show that for the concrete specimen and the steel member, which are two kinds of structures with clear angles, the detection and matching of their characteristic points in the test are relatively smooth, and the error between the node data obtained in

the point cloud model and the standard data can be controlled within 2.5%. In addition, automated modeling is simpler and faster than the traditional method, with the average modeling time for a single specimen not exceeding 1.5 h.

This benchmark study not only confirms the reliability of the method on simple structures but also demonstrates its scalability for modeling complex transmission towers (Section 4). The quantitative accuracy and efficiency metrics provide a solid foundation for field applications.

4. Field-Applicable Validation: Scaled Transmission Tower Modeling via UAV Photogrammetry

4.1. Experimental Overview

To verify the feasibility of the proposed innovative FEM method, this study conducted an outdoor field test on a “transmission tower model.” The dimensions and structure of the transmission tower model are shown in Figure 8. In the experiment, FEMs of the target tower were constructed using both point cloud reconstruction technology and traditional engineering drawings. The natural frequencies and mode shapes of the structure were selected as FEA results for comparative analysis.

Firstly, throughout the experiment, the main image acquisition device used was a DJI UAV, model Matrice M300 RTK (Da-Jiang Innovations, Shenzhen, China), equipped with a Saier 6100 gimbal camera and a 40MM professional surveying lens. The camera’s performance was comparable to that of a standard smartphone, with the camera model SHARE6100PLUS061X, an aperture of $f/0$, ISO sensitivity of ISO-250, and a fixed focal length of 40 mm. The final resolution of the captured raw image data was 9504×6336 pixels, and the EXIF file version for the automatic reading of camera internal parameters by the computer was 0231. In principle, the image dataset used for generating the point cloud model with computer vision technology needs to meet the following requirements: (1) capture a large number of photos from multiple angles; (2) ensure the lens captures the entire structure’s outline; (3) maintain clear and visible images; (4) ensure at least 50% overlap between images in the raw dataset to guarantee accuracy in subsequent feature matching as well as the continuity and completeness of the point cloud model; and (5) minimize direct sunlight exposure in the collection environment to reduce lighting-related errors in the raw images. However, based on the structural characteristics of transmission towers and to simulate the entire process of obtaining image datasets of transmission towers in real engineering applications, this study was conducted in a relatively idealized environment. A UAV equipped with a camera lens was used to orbit the transmission tower model and capture video. The video file was then imported into VLC media player, where frames of interest were extracted according to research needs. These frames were subsequently cropped individually in Krita 5.2.0 software and saved in JPG format, forming the raw image dataset required to build the point cloud model, which was then processed in the computer for subsequent steps. In this study, the use of a UAV ensured that the camera lens could capture all key nodes of the transmission tower model comprehensively during the shooting process, supporting subsequent FEM. This approach also aligns with the advantages of using computer vision in real engineering applications, such as “long-range” and “non-contact” capabilities.

Subsequently, based on the principles and methods described above, the sparse point cloud reconstruction, depth information acquisition, and dense point cloud reconstruction of the transmission tower model were completed sequentially. After the dense point cloud reconstruction is completed, the dense point cloud model can be saved in PLY file format, exported as a Poisson point cloud model or Delaunay point cloud model, and viewed in point cloud processing software such as Meshlab 2023.12 to assess the reconstruction re-

sults. According to the needs of the experimental study, each part of the transmission tower model is examined to determine whether there are insufficient points. In such cases, additional camera poses are added in the corresponding areas to reestablish the original image dataset, and the above steps are repeated to ensure the completeness of the point cloud model reconstruction.

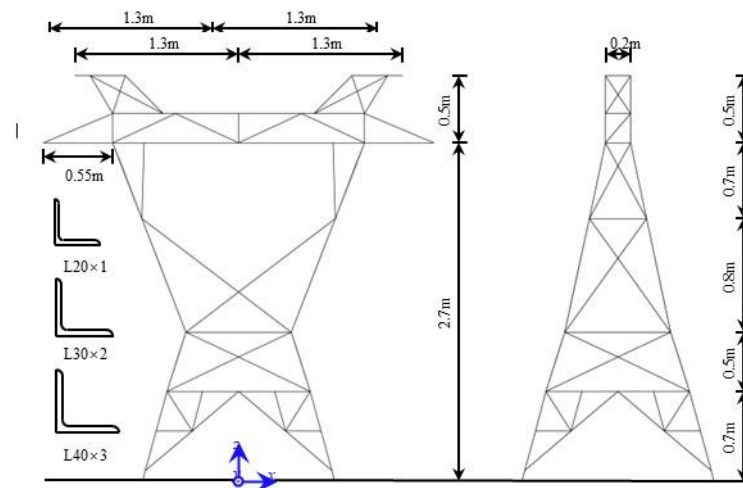


Figure 8. Dimensions and construction of transmission tower models.

Finally, the developed APDL command stream generator was utilized to assign material properties and boundary conditions to the model, automatically generating executable command streams for ANSYS. These commands were then input into the software to perform FEA, thereby completing the entire workflow. It should be noted that the nodal coordinates in the virtual coordinate system of the point cloud model do not represent actual physical dimensions, and the scale of the model differs significantly from the real structure. To establish a full-scale finite element model, coordinate transformation from the virtual to the real-world coordinate system must be performed. This conversion is achieved by calculating a scaling factor based on the distance ratio between corresponding point pairs in both coordinate systems. In the virtual coordinate system, the distance between two points is calculated as shown in Equation (6):

$$l_{ab} = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2 + (z_a - z_b)^2} \quad (6)$$

In the above equation, l_{ab} represents the distance between points A and B in the virtual coordinate system, where x_a , y_a and z_a are the x , y , and z coordinates of point A, and x_b , y_b and z_b are the x , y , and z coordinates of point B, respectively. In the real coordinate system, the actual distance between the corresponding point pairs on the transmission tower model is measured manually, typically using an engineering tape measure to obtain the distance l_{ab}' . The ratio between the two distances is then used to calculate the scaling factor, as shown in Equation (7):

$$s = \frac{l_{ab}}{l_{ab}'} \quad (7)$$

Consequently, this study may inevitably encounter human errors or computational errors in distance measurement and scaling factor calculation. However, the impact of such minor errors on the final results falls within the acceptable range in engineering.

4.2. Analysis of Experimental Results

4.2.1. Original Image Dataset Collection

Based on the image dataset collection devices and considerations mentioned in the overview of the experiment, this study captured images of the transmission tower model in the university's testing facility. Image data collection for this experiment was conducted on a sunny day between 4 PM and 5 PM, ensuring good and even lighting while avoiding the effects of harsh midday sunlight. To ensure that images of the transmission tower model are captured from multiple viewpoints, the UAV was controlled to orbit the model at a distance of 2 m and at various heights to obtain the corresponding video files. The arrangement of the transmission tower model and the UAV, along with other experimental devices, is shown in Figure 9.

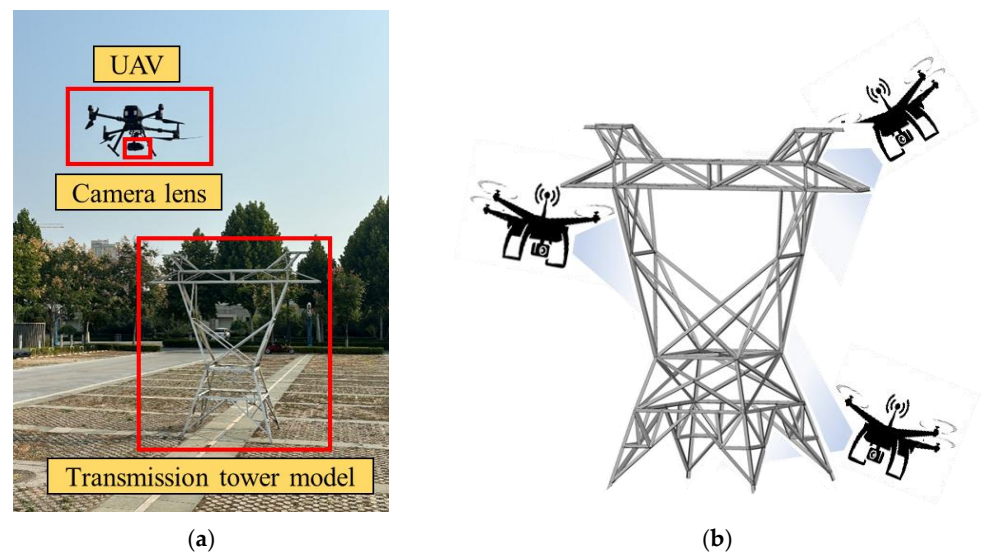


Figure 9. Setup for image dataset collection experiment: (a) field photograph of experiment; (b) experimental schematic.

After the UAV captures the video files, it is essential to ensure that a sufficient number of images are obtained during the video extraction process in VLC Media Player, covering every angle and detail of the structure while discarding frames that exhibit motion blur due to sudden movements of the UAV. Generally speaking, the more images collected, the better the reconstruction of the point cloud. In this experiment, a total of 172 JPG image files were extracted. However, to ensure the accuracy and completeness of the final generated point cloud model, the extracted images must be processed in Krita and MATLAB. In Krita, the images are primarily cropped along the contours of the transmission tower model, and the cropped images are then saved as JPG files. This approach aims to accelerate the subsequent generation of the point cloud model by the computer, reducing unnecessary feature detection and matching in the images.

4.2.2. Model Tower Point Cloud Model Reconstruction

After obtaining the raw image dataset, the principles mentioned above can be applied to the sparse point cloud reconstruction, depth information acquisition, dense point cloud reconstruction, and FEM of the transmission tower model. The reconstruction results of the sparse point cloud model and the dense point cloud model are shown in Figure 10.

Firstly, the unordered image set is input into the program for feature extraction, feature detection, and feature matching. After a short wait, the sparse point cloud model of the target transmission tower and the camera poses of the UAV in the shooting scene are

obtained, as shown in Figure 10a, which illustrates a preliminary shape of the transmission tower. In this model, the various parts do not contain enough points; however, the overall outline of the transmission tower model is depicted based on the sparse feature points on the structure's surface. In Figure 10a, the orange tetrahedral shapes represent the camera poses during the UAV's filming process, corresponding to the viewpoints from which each original image dataset was captured, including the camera's position coordinates and shooting angles in the virtual coordinate system. Combining Figure 10a with the methods mentioned earlier, the resulting sparse point cloud model and camera poses, after undergoing incremental reconstruction, triangulation, and iterative optimization, are largely consistent with the actual experimental conditions. During this process, it is vividly observed that the sparse point cloud model becomes more complete with the addition of camera poses, exemplifying incremental reconstruction. The computer does not take too long to process feature points and generates a feature matching matrix based on the results of feature matching, indicating whether the matching level is satisfactory. The feature point matching matrix for the transmission tower model in this experiment is shown in Figure 11.

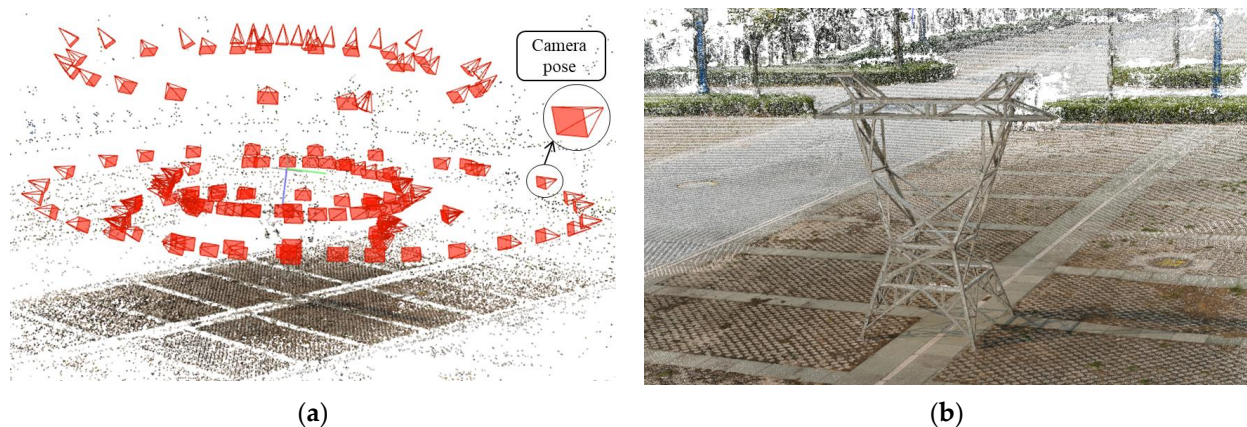


Figure 10. Reconstruction results of point cloud models: (a) sparse point cloud model; (b) dense point cloud model.

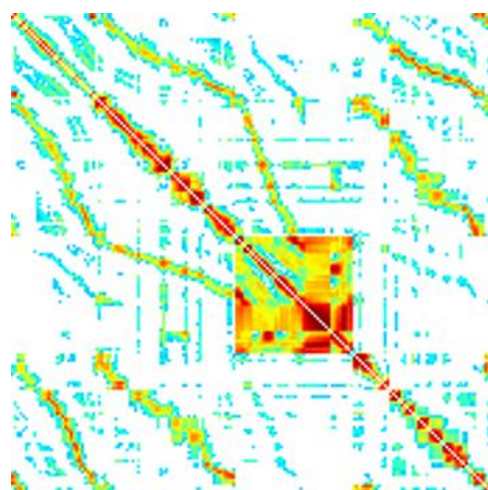
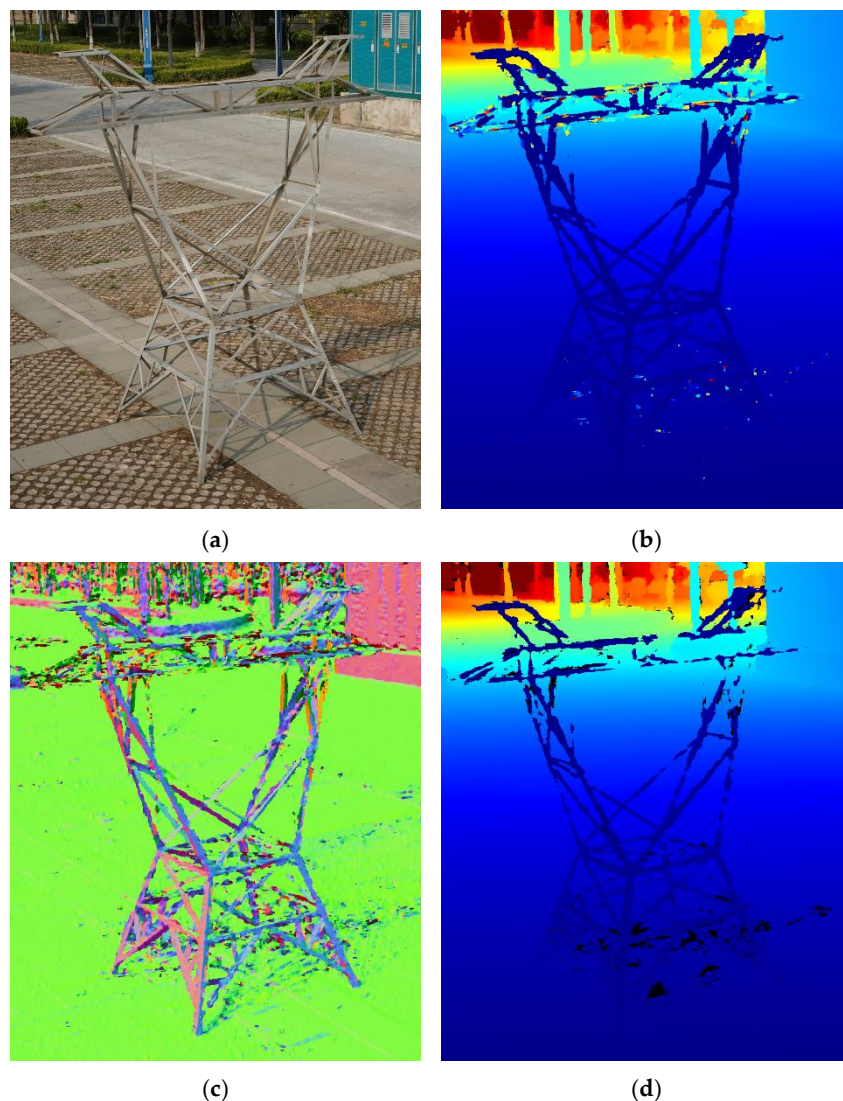


Figure 11. Feature point matching matrix.

The feature point matching matrix allows for the visualization of a heatmap of the matching results. The distribution of this matching matrix shows relatively dense regions, indicating a rich presence of feature points in the original data and sufficient matching between images. The dense color near the diagonal in the figure implies a higher number

of matching points between adjacent images, indicating good coherence in camera movement and minimal variation in shooting angles. This suggests that during the data acquisition process, the control of the camera's movement was relatively stable, which is beneficial for the continuity and completeness of the target point cloud during 3D reconstruction. However, the parallel color bands in the figure exhibit a loose distribution in some areas, with certain regions showing insufficient tightness in the arrangement of the color bands. This may indicate that in some areas of data collection, due to the instability of the UAV's flight, there was a lack of rigor in controlling the camera's movement, resulting in insufficient matching between specific images. Considering the above analysis of the figure, it can be concluded that, overall, the sparse point cloud reconstruction process made effective use of the information from the original image set.

After obtaining the sparse point cloud model of the transmission tower structure and the camera pose information, the reconstruction model is incomplete, making it difficult to obtain the coordinates of the key nodes of the tower model. To further construct the dense point cloud model of the target structure, depth information estimation must be performed on the original image set based on the principles mentioned earlier. An example of the depth estimation results in this experiment is shown in Figure 12. Figure 12a displays the original image, while Figure 12b and 12c present the depth map and normal vector map based on "photometric consistency," respectively. Figure 12d and 12e show the depth map and normal vector map based on "geometric consistency," respectively.





(e)

Figure 12. Image processing results: (a) original image; (b) photometric depth map; (c) photometric normal vector map; (d) geometric depth map; (e) geometric normal vector map.

The depth map distinguishes the depth of the target from the camera plane based on variations in RGB color. In image processing, the normal vector map provides a more intuitive assessment of the 3D fitting accuracy of the target structure. By analyzing Figure 12, it can be observed that the depth map generated based on “photometric consistency” displays relatively chaotic depth information, particularly in areas with structural details, where color differentiation is not clear, indicating low accuracy in depth estimation. Furthermore, normal vector mapping indicates the presence of roughness on the model surface, with a high number of non-target points, resulting in suboptimal surface fitting. However, in the depth map optimized based on “geometric consistency,” the color differentiation of depth information is significantly improved, especially in the key structural parts of the tower, where the depth estimation is closer to the actual situation, indicating that geometric optimization effectively enhances the accuracy of depth estimation. Simultaneously, normal vector mapping filters out most non-target points, resulting in a smoother surface of the tower that is closer to the actual structure. This demonstrates that through the joint optimization of photometric and geometric consistency, the generated depth map can better represent the actual structure of the target building, providing a more reliable foundation for subsequent 3D reconstruction. Additionally, this optimization process can reduce reconstruction time, enhance the accuracy and continuity of point cloud construction, and yield more precise reconstruction results.

Subsequently, based on the obtained depth information and the principles described in this paper, the sparse point cloud model was filled, resulting in the final dense point cloud model of the transmission tower.

As shown in Figure 10b, the 3D model displayed in the virtual coordinate system partially restores the target structure. Despite using UAV orbiting photography to achieve a more detailed reconstruction of the transmission tower model, some areas still exhibit certain deficiencies, indicating that there is room for improvement in the completeness of the dense point cloud model. In terms of the experiment itself, the constructed dense point cloud model essentially met the experimental requirements, with the structure’s proportions and color textures closely resembling the actual tower. Furthermore, it is evident that the number of points in the dense point cloud model is significantly greater than that in the sparse point cloud model (see Table 1 for detailed experimental data). More importantly, this model is now capable of fulfilling the function of obtaining the coordinate information of key nodes for the FEM in practical applications.

Table 1. Image-based point cloud generation metrics for structural specimens.

Target Structure	Number of Images	Sparse Point Cloud Points	Dense Point Cloud Points
C30 Concrete Specimen	32	14,312	1,144,374
Q235B Steel Element	53	14,810	912,880
Scaled Tower Model	173	100,262	4,914,095

4.2.3. Experimental Comparison of Finite Element Modeling Methods for Transmission Towers Based on Point Cloud Techniques and SmartTower Finite Element Modeling Methods

Based on the generated high-precision point cloud model, the finite element model of the transmission tower was successfully constructed in this study using the aforementioned method. The node information after coordinate conversion was inputted into the developed automated APDL command flow generator, and the nodes were connected in an orderly manner to form the model framework. Appropriate material properties and boundary conditions were then assigned. Specifically, the model uses a material with an elastic modulus of 69 GPa, Poisson's ratio of 0.3, and density of 3000 kg/m³—representative of an aluminum alloy or similar lightweight structural material. Three cross-sectional types were defined, corresponding to L-shaped angle sections of different dimensions: Section 1 (L40 × 40 × 3), Section 2 (L20 × 20 × 1), and Section 3 (L30 × 30 × 2). The finite element model employs BEAM188 and BEAM189 elements for discretization, with each structural member typically represented by a single element. Fixed boundary conditions are applied at four key support points (keypoints 7000, 7001, 7002, 7003) to simulate rigid supports. The final output is a complete APDL command flow that can be directly imported into ANSYS software. The command stream can be analyzed by the finite element software for subsequent mechanical analysis, thus realizing the complete transformation from a 3D point cloud model to a computable mechanical model.

To validate the engineering applicability of this UAV point cloud-based automated FEM system, a comparative analysis was conducted against SmartTower V2024—a commercial FEM software specialized for transmission towers and widely adopted in power industry design practices. The comparative results are summarized as follows:

- (1) Nodal Coordinate Accuracy: Both methods generated ANSYS APDL command streams containing complete nodal coordinates of the FEM. As presented in Table 2, 10 node coordinates were arbitrarily selected for comparison, and the maximum coordinate deviation between the point cloud-based model and the reference SmartTower model was limited to 10%, demonstrating high geometric fidelity.

Table 2. Comparison of node coordinate accuracy.

No.	Point Cloud Finite Element Modeling	SmartTower Finite Element Model	Maximum Error
1	(0.4142, 0.4147, −0.0319)	(0.4200, 0.4200, 0.0000)	7.6%
2	(−0.4351, 0.4045, 0.0034)	(−0.4200, 0.4200, 0.0000)	4.9%
3	(0.4094, −0.4231, 0.0099)	(0.4200, −0.4200, 0.0000)	3.1%
4	(−0.4360, −0.4218, −0.0469)	(−0.4200, −0.4200, 0.0000)	3.8%
5	(0.9193, 0.1224, −1.7890)	(1.0000, 0.1150, −1.7300)	9.2%
6	(−1.0817, 0.0365, −1.6984)	(−1.0000, 0.1150, −1.7300)	8.1%
7	(0.9220, 0.1044, −1.7808)	(1.0000, −0.1150, −1.7300)	9.2%
8	(−1.0597, −0.1190, −1.7028)	(−1.0000, −0.1150, −1.7300)	5.9%
9	(0.9278, 0.0382, −1.5535)	(1.0000, 0.1150, −1.5000)	7.2%
10	(−1.0619, 0.0491, −1.4779)	(−1.0000, 0.1150, −1.5000)	6.1%

- (2) **Modeling Efficiency:** As presented in Table 3, the proposed point cloud-based methodology significantly improves efficiency by reducing the total modeling time by 88% (from 1.5 days to 4.3 h) compared to SmartTower’s workflow using engineering drawings.

Table 3. Comparative analysis of modeling efficiency metrics.

FEM Approach	Time
Point cloud	4.3 h
SmartTower	1.5 days

- (3) **Modal Analysis Performance:** Focusing on SHM requirements, the first three natural frequencies and corresponding mode shapes were compared (Table 4 and Figure 13). While minor discrepancies existed between the two models (maximum frequency error: 3.69% for the first mode), all differences fell well within the acceptable 5% threshold for civil engineering monitoring.

Table 4. Comparison of natural frequency calculations for first three vibration modes.

FEM Approach	First-Order Frequency	Second-Order Frequency	Third-Order Frequency
Point cloud	15.90	35.30	43.34
SmartTower	16.51	35.16	43.46
Relative error	3.69%	0.39%	0.27%

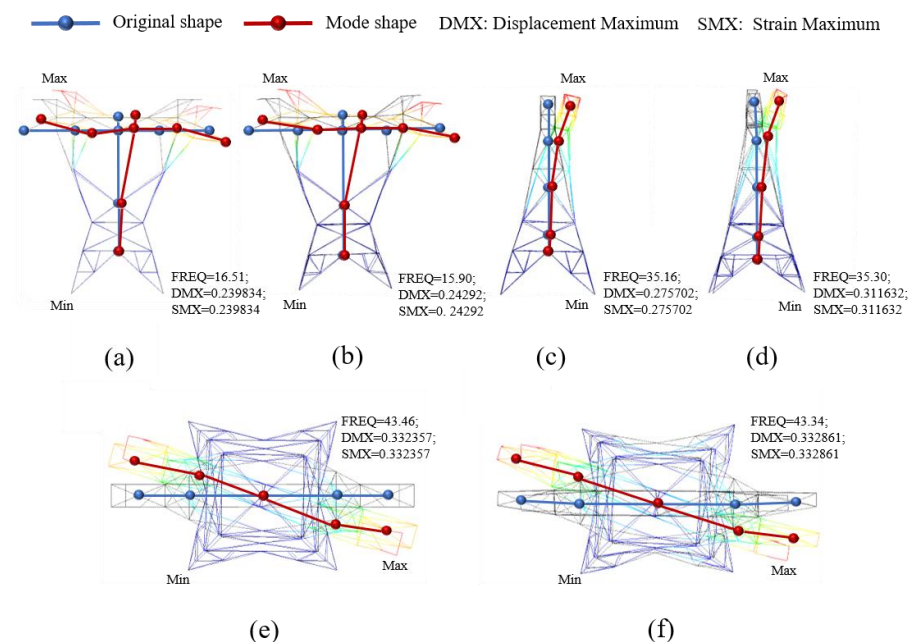


Figure 13. Calculation results of first three orders of vibration patterns of two finite element models: (a,c,e): calculation results of finite element model constructed by using engineering drawings; (b,d,f): calculation results of finite element model constructed by using point cloud model.

Notably, the mode shapes exhibited identical deformation patterns (Figure 13), confirming the kinematic consistency of the proposed method. These results validate that the modeling accuracy of the point cloud technique used in this paper basically meets the traditional modeling standards.

It can be seen that this approach shows promising prospects for the real-time monitoring of transmission towers in engineering and holds significant potential in the field of non-contact health monitoring. This method is expected to serve as a bridge between com-

puter vision and structural FEA, enhancing flexibility and robustness in practical engineering operations within the field of SHM while also facilitating ease of use.

5. Conclusions

This study presents an innovative framework for the automated finite element modeling of transmission towers through UAV-based point cloud reconstruction. The key contributions and findings can be summarized as follows:

- (1) **Methodological Innovation:** The proposed pipeline from computer vision to finite element modeling establishes an automated workflow that connects UAV photogrammetry (90% feature coverage) to structural analysis. Multi-view stereo matching, geometric photometric refinement, and the development of an automated APDL command generator allow for the conversion of dense point clouds into finite element models, filling an important research gap in non-contact structural modeling. It should be noted that research at this intersection is currently very limited, and direct quantitative comparison with other methods is not yet feasible.
- (2) **Technical Validation:** This study systematically validated the effectiveness and feasibility of the proposed method through benchmark tests on concrete specimens and steel structural components, as well as field tests on a scaled transmission tower model. The experimental results demonstrate that the method achieves high modeling accuracy for target structures. Compared with traditional FEM software (e.g., SmartTower), the proposed approach maintains nodal coordinate errors within 10%, limits frequency errors in modal analysis to 3.7%, and shows excellent consistency in mode shapes. Remarkably, it improves modeling efficiency by 88%, reducing the conventional 1.5-day workflow to under 5 h. These results meet industry standards for civil engineering monitoring accuracy, confirming the reliability of the technical approach.
- (3) **Engineering Significance:** This approach solves two core pain points in transmission tower modeling through point cloud technology: first, it breaks through the dependence of traditional methods on engineering drawings and provides a feasible digital modeling solution for the old towers that lack complete drawings; second, it shortens the modeling cycle time for a single tower, which significantly improves the efficiency of large-scale inspections of the power grid. This “computer vision + finite element” fusion paradigm provides a new technical path for the rapid safety assessment of electric power infrastructure, which is especially suitable for engineering scenarios such as post-disaster emergency assessment and tower monitoring in remote areas.
- (4) **Limitations and Future Work:** Current limitations include lighting-dependent reconstruction accuracy and predefined material assumptions. The current method relies on a predefined database for cross-sectional properties and does not automatically identify member geometries from point clouds. Structural defect recognition (e.g., bolt loosening, member deformation) is also not addressed in this study. Future work will explore automated cross-section recognition through point cloud geometric analysis or deep learning-based classification, as well as the integration of defect detection techniques into the point-cloud-to-FEM pipeline. Future work will also integrate LiDAR–thermal fusion and deep learning for robust automated modeling, particularly for aged infrastructure and tower-line systems.

Overall, this approach provides insights and methodologies for the application of computer vision to structural FEA and serves as a reference for developing software related to FEM using computer vision technology. Improving modeling accuracy and automation is a focus and direction for future work in this research.

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